



BUDDY: AN INTERACTIVE COMPANION ROBOT FOR FAMILY

MISS THANANYA	OSOCHPROMMA
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**A PROJECT REPORT SUBMITTED IN PARTIAL
FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE
BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING**

**FACULTY OF ENGINEERING & INTERNATIONAL COLLEGE
MAHIDOL UNIVERSITY**

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Computer Engineering Project
entitled
BUDDY: AN INTERACTIVE COMPANION ROBOT FOR FAMILY

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18 หน้า

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DATE OF GRADUATION	July 15, 2024

ABSTRACT

An abstract (250-400 words) should provide a concise summary of your entire thesis. It should report significant elements of your thesis including background or introduction in brief, objectives, statistical data, key findings, and conclusion. Mathematical formulas, diagrams, and other illustrativematerials are not recommended for inclusion. A strong abstract should be self-contained; without abbreviations, footnotes, references. Outside readers typically view the abstract before deciding to read the thesis, so it should be well written, logical, and a complete reflection on you work. The abstract is typically written last after finishing chapter 4 and 5.

KEYWORDS : MLP

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CHAPTER 1

INTRODUCTION

1.1 Background

Background Loneliness and social isolation are rising, particularly among the elderly population. It is concerning as Thailand is entering an aged society, a community in which the percentage of people 60 and older makes up 20% or more of the total population in a certain area, or when the percentage of people 65 and older makes up 14% or more of the total population in the same location [1]. According to the National Statistical Office Ministry of Digital Economy and Society, as of 2024 (the latest update at the moment), the elderly population has risen to 20% [2]. There are ways to take care of elders when the main caretaker can not be there as much as they want, like taking them to a nursing home or hiring a caretaker. However, most of the elders prefer to stay home, and it is hard to find people who are trustworthy for your loved one. Elder people might not be enthusiastic about using a smartphone, it is even harder when their caretakers have to go outside, and there is no way to communicate with elders. Implementing IoT in the household might be a choice, for instance, you can buy a security camera to monitor them. Most security cameras indeed allow you to do two-way communication, but it is not ideal, as it contains a lot of feedback. Moreover, to allow them to also communicate with you, you have to implement more things. According to the provided information, we think that it would be better if the elders could be in their own homes with a companion that helps monitor them, and let them connect with their caretakers by voice or an easy-to-access button.

1.2 Objective

1. To introduce a video call function .
2. To develop a emotion progress report
3. To enhance Human-Robot Interaction (HRI)

CHAPTER 2

LITERATURE REVIEW

Technology has become an essential tool in addressing the challenges associated with an aging population. One area of growing interest is the development of smart companion systems aimed at supporting elderly individuals in their daily lives. These systems integrate artificial intelligence and communication platforms to enhance quality of life, improve safety, and foster independence. This research aims to provide personalized and accessible care for elderly individuals and their families, supporting their emotional well-being. This literature review investigates the application of each selected study to the development of a smart companion system for elderly care. The primary goals of the study are to implement a function to predict emotions based on audio and facial recognition data, conduct natural conversations, introduce a video call function for real-time communication, and develop a monitoring system that functions as a camera for elderly care. These features aim to meet both the emotional and practical needs of elderly individuals, offering comprehensive support. The scope of this research is shaped by the areas explored in the selected studies. The Multimodal Emotion Recognition Using Feature Fusion: An LLM-Based Approach focuses on emotion recognition using multimodal inputs by converting audio and visual features into structured text, offering a new approach for LLM-based classification accuracy. The AI Affective Conversational Robot with Hybrid Generative-Based and Retrieval-Based Dialogue Models investigates emotionally-aware dialogue systems through deep learning-based sentiment analysis and dialogue generation. The Development of an Intelligent Dialogue Agent for Older Adults explores how reinforcement learning techniques can adapt conversation topics based on user preferences to promote social companionship. These studies collectively provide insights into multimodal emotion recognition, sentiment prediction, and personalized dialogue generation, forming the foundation for this research.

2.1 Multimodal Emotion Recognition Using Feature Fusion: An LLM-Based Approach [1]

Multimodal emotion recognition has evolved from traditional machine learning methods to deep learning-based multimodal fusion techniques, integrating speech, facial expressions, and textual data. While deep learning models like CNNs, RNNs, and transformers have improved classification accuracy, they often suffer from high computational costs and feature alignment challenges. Recent advancements in Large Language Models (LLMs) such as BERT, RoBERTa, and GPT-3 have enabled emotion recognition through textual representations of multimodal inputs. This paper addresses key gaps by proposing a novel feature fusion approach that converts audio and visual features into structured text, processed by a fine-tuned DistilRoBERTa model for emotion classification. The method achieves state-of-the-art accuracy on benchmark datasets, demonstrating that LLM-based textual fusion can match or exceed traditional multimodal deep learning models while improving scalability and interpretability. This approach bridges the gap between multimodal emotion recognition and NLP, offering a promising direction for future affective computing applications.

2.2 AI Affective Conversational Robot with Hybrid Generative-Based and Retrieval-Based Dialogue Models [2]

ChatBot technology has become a widely used in various application fields. An important topic in the research on conversational robots is the improvement of their temperature during operation for enhanced user interaction. In this study, we propose an artificial intelligence affective conversational robot (AIACR), which is an integration of an artificial intelligence deep learning sentiment analysis model and generative-and retrieval-based dialogue models. The sentiment analysis model developed in this study uses three models, namely, multilayer perceptron (MLP), long short-term memory (LSTM) and bidirectional long short-term memory (BiLSTM). Moreover, word2vec and semantics are utilized as the basis for similarity ranking models. The deep learning dialogue model, sentiment analysis model, and similarity model were integrated and compared as well. The experimental results show that the sentiment analysis model, similarity model, and dialogue model respectively utilize BiLSTM, word2vec, and the retrieval-based model to achieve the best dialogue performance. The major research contributions of this study are the developed AIACR and the proposed affective conversational robot index (ACR Index) as a criterion for evaluating the effectiveness of emotional dialogue robots.

The main objectives of this paper are as follows:

- To compare the effects of conversational robots in retrieval- and generative-based models.
- To use artificial intelligence (AI) technology such as multilayer perceptron (MLP), long short-term memory (LSTM), and bidirectional LSTM (BiLSTM) to conduct sentiment analysis comparison.
- To integrate the dialogue model with the sentiment prediction model to produce a dialogue with emotions.

2.3 Development of an Intelligent Dialogue Agent for Older Adults : Evaluation of Functions to Control Spontaneous Talk and Coordinate Speech Content [3]

A Reinforcement Learning-Based Approach Intelligent Dialogue Agents (IDAs) have been increasingly utilized to support elderly care by offering social companionship and promoting preventive care activities. Traditional voice-interaction devices like Google Home and Amazon Echo provide information in response to user queries but cannot initiate spontaneous conversations. Recent research has focused on developing IDAs with spontaneous talking functions (STF) to foster more natural and friendly interactions, enhancing users' sense of familiarity and engagement. Kitakoshi et al. (2019) proposed an IDA that combines STF with a Speech-Content Coordinating Function (SCCF) to adapt speech content based on user preferences and behavioral trends. The SCCF employs reinforcement learning (RL) algorithms, specifically a policy gradient method, to select appropriate speech topics at the right moments. Experiments conducted with students demonstrated that users responded positively to personalized content but highlighted the need to optimize speech length and RL parameters to improve learning performance. The study suggests that integrating adaptive dialogue generation with preventive care systems can encourage habitual use among older adults, contributing to comprehensive preventive care frameworks.

CHAPTER 3

METHODOLOGY

This section describes the methodology for developing the smart companion system for elderly care. The methodology comprises system design, data collection, pre-processing techniques, machine learning methods, hyperparameter configurations, and performance evaluation. The technical methods used in this study include speech recognition, natural language processing, sentiment analysis, and WebRTC-based intercommunication. Each stage of the methodology is explained to ensure clarity and reproducibility.

3.1 Caretaker Application

3.1.1 UI/UX Design

As the primary user is the caretaker who needs quick access to important information, therefore this application was designed with a focus on simplicity, and usability. The design process began with wireframing with Figma, where the layout, visual hierarchy, and user flow were all maintained by outlining each page.

Key Components:

- Hardware (Lightweight Client): Captures speech and visual data.
- Mobile Application: Allows monitoring and communication with hardware users.
- Speech Recognition: Google Speech to Text API for Thai speech recognition.
- API Server: FastAPI server to handle tokenization and requests.
- Local LLM: GEMMA 3 for emotion classification and dialogue generation.
- Text-to-Speech: Google Text to Speech API converts the model responses into audio feedback.
- WebRTC Communication: Supports video calls between hardware users and software users.
- Database Systems: PostgreSQL for user information and MongoDB/Pinecone

3.2 Data

The system relies on audio and visual data collected from elderly users during natural conversations. Audio data is captured using microphones embedded in the hardware client, while visual data is captured through cameras. The collected data is processed for emotion recognition and conversation generation.

Data Pre-Processing

- Audio Tokenization: Google Speech to Text API performs automatic speech recognition and converts speech into text.
- Text Tokenization: User inputs are tokenized and filtered before being sent to the LLM.

3.3 Methods

Speech Recognition - Google Speech to Text API

Google Speech-to-Text API is a cloud-based automatic speech recognition (ASR) service that converts spoken language into written text. It supports Thai speech recognition using deep learning models and accommodates various accents and noise conditions. The API processes audio through the following steps:

- Audio Ingestion: Accepts audio input from live streaming or pre-recorded files in various formats.
- Feature Extraction: Processes audio waveforms and extracts relevant features using deep learning techniques.
- Model Inference: Uses Google's neural network models trained on large-scale Thai speech data to generate transcriptions.
- Post-Processing: Applies linguistic and contextual adjustments to improve accuracy, including punctuation and formatting.
- Text Output: Returns transcribed text along with confidence scores and optional speaker diarization.

Emotion Recognition - GEMMA 3

- Audio Emotion Recognition: GEMMA 3 model classifies the emotion from the converted texts.

Dialogue Generation - GEMMA 3

- The GEMMA 3 model generates emotionally-aware conversations by combining sentiment analysis and dialogue generation.

Video Call Communication - WebRTC

- WebRTC is used to facilitate real-time video communication between elderly users and their family members.

Mood Summary Generation

- Mood Summary is summarized based on conversation data using GEMMA 3 and stored in MongoDB/Pinecone.

3.4 Database Schema

The system utilizes PostgreSQL for user-related data and MongoDB/Pinecone for emotion summaries and conversation as shown in Figure 3.2. The key database tables and collections are:

PostgreSQL Database (User Data)

- **Users Table**
 - user_id (Primary Key, INT)
 - name (VARCHAR)
 - age (INT)
 - email (VARCHAR)

MongoDB (Emotion Summary Storage)

- **Emotion Summaries Collection**
 - _id (Primary Key, ObjectId)
 - user_id (Foreign Key, INT, references **Users(user_id)** in PostgreSQL)
 - timestamp (TIMESTAMP)
 - mood_summary (TEXT)

MongoDB (Dialog History Storage)

- **Dialog History Collection**
 - _id (Primary Key, ObjectId)
 - user_id (Foreign Key, INT, references **Users(user_id)** in PostgreSQL)
 - timestamp (TIMESTAMP)

- message (TEXT)
- speaker (VARCHAR, "user" or "AI agent")
- session_id (VARCHAR)
- detected_emotion (VARCHAR)

Pinecone (Dialog Embedding Storage)

- **Dialog Embedding Collection**

- _id (Primary Key, UUID)
- dialog_id (Foreign Key, ObjectId, references **Dialog History(_id)** in MongoDB)
- embedding (Vector)
- metadata (JSON)

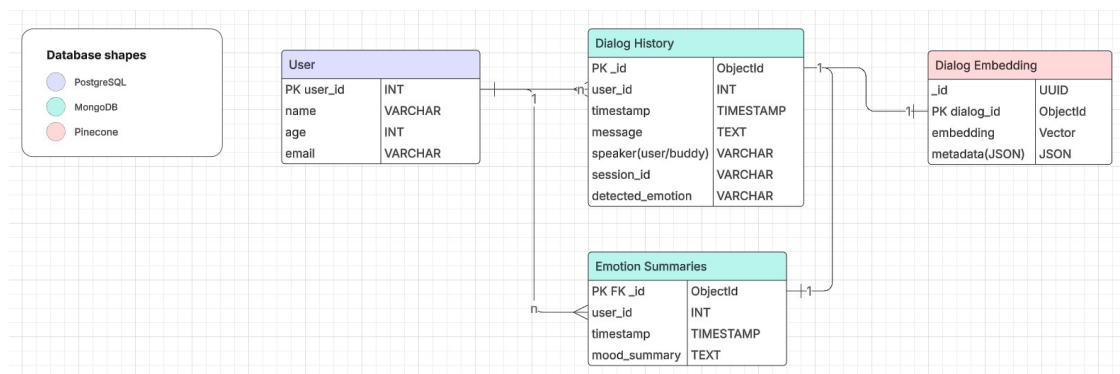


Figure 3.1 ER-Diagram for the Database

3.5 Hyperparameter Configurations

- Google Speech-to-Text API:
 - Model: `latest\_long` (optimized for long-form Thai speech)
 - Confidence Threshold: 0.7 (filters low-confidence transcriptions)
 - Enable Automatic Punctuation: `True`
 - Audio Encoding: `LINEAR16`
 - Sample Rate: 16000 Hz
- GEMMA 3:
 - Model: `gemma:3b` (3-billion parameter version)
 - Max Tokens: `4096` (limits response length)
 - Temperature: `0.8` (higher values increase randomness)

- Top-k: `40` (limits token selection for coherent responses)
- Top-p: `0.9` (nucleus sampling for diversity)
- Repeat Penalty: `1.2` (reduces repetitive text)
- Seed: `None` (random generation unless specified)
- GPU Acceleration: `Enabled` (if running on a supported GPU)
- WebRTC Connection:
 - STUN/TURN Servers: Configured for NAT traversal
 - Video Codec: `VP8`
 - Audio Codec: `Opus`
 - Bitrate: Adaptive
 - Latency Mode: Low-latency enabled

3.6 Performance Evaluation

The performance of the smart companion system will be evaluated based on the following metrics:

3.6.1 Emotion Recognition Accuracy

Emotion recognition accuracy will be measured using standard classification evaluation metrics:

- **Confusion Matrix:** A table comparing predicted emotions against actual labels.

It consists of:

- **True Positive (TP):** Correctly predicted positive instances.
- **False Positive (FP):** Incorrectly predicted positive instances.
- **True Negative (TN):** Correctly predicted negative instances.
- **False Negative (FN):** Incorrectly predicted negative instances.

- **Precision, Recall, and F1-Score:**

- **Precision** = $\frac{TP}{TP+FP}$

- Measures the proportion of correctly predicted positive instances among all predicted positive instances.

- **Recall** = $\frac{TP}{TP+FN}$

- Measures the proportion of correctly predicted positive instances among

all actual positive instances.

- **F1-Score** = $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
- The harmonic mean of precision and recall, balancing both metrics.

- **Overall Accuracy:**

- Accuracy = $\frac{TP+TN}{TP+TN+FP+FN}$
- Measures the overall correctness of the model by considering both positive and negative instances.

3.6.2 Answer Generation Quality

The evaluation of the model's ability to generate coherent and relevant responses will be performed using the following metrics:

- **ROUGE Score (Recall-Oriented Understudy for Gisting Evaluation):**

- Evaluates recall-based overlap of n-grams and longest common subsequence.
- ROUGE-N = $\frac{\sum_{s \in \text{ReferenceSummaries}} \sum_{gram \in s} \text{CountMatch}(gram)}{\sum_{s \in \text{ReferenceSummaries}} \sum_{gram \in s} \text{Count}(gram)}$

- **Exact Match (EM):**

- Measures the percentage of cases where the generated response matches the reference response exactly.
- EM = $\frac{\text{Number of Exact Matches}}{\text{Total Samples}}$

- **Perplexity (PPL):**

- Computes the likelihood of the generated response under the model's probability distribution, indicating fluency.
- PPL = $e^{(-\frac{1}{N} \sum_{i=1}^N \log P(w_i))}$
- Where:
 - * n is the number of words (or tokens) in the generated response.
 - * $P(w_i)$ is the probability of the i -th word w_i in the generated response according to the model's distribution.

- **Human Evaluation:**

- Users assess coherence, fluency, and emotional alignment through surveys.

3.6.3 Speech-to-Text (STT) Performance

The accuracy of the Thai speech-to-text module (e.g., Google Speech to Text API) will be evaluated using:

- **Word Error Rate (WER):**

- $WER = \frac{S+D+I}{N}$

- where S = substitutions, D = deletions, I = insertions, and N = total words in reference text.

- **Character Error Rate (CER):**

- $CER = \frac{S+D+I}{N}$

- where N represents total characters in reference text, useful for Thai language without explicit word boundaries.

- **Human Evaluation:**

- Assessing fluency, tone accuracy, and named entity recognition (NER) for Thai names and locations.

3.6.4 Video Call Quality

Measured by:

- **Latency:** Delay between user input and system response.
- **Jitter:** Variability in packet arrival times.
- **Packet Loss:** Percentage of lost data packets.

3.6.5 Software

- Assessed using Black box test and then White box test.
- Utilized Selenium as Software Application Testing Tools.

CHAPTER 4

RESULT

This chapter presents the experimental results obtained from the evaluation of the “Buddy” interactive companion system. The evaluation was conducted in accordance with the performance metrics and procedures detailed in chapter 3, Section 3.6, and modified to incorporate several evaluation techniques.

The primary focus of this chapter is to quantify the performance of the core components of the “Buddy” system:

- **Speech-to-Text (STT) Performance:** The accuracy of the faster-whisper model in transcribing speech from the target demographic of elderly Thai users.
- **Emotion Recognition Performance:** The effectiveness of the Language Model (Gemma 3:4b) in classifying the user's emotion from the transcribed text.
- **Dialogue Generation Quality:** The coherence, fluency, and appropriateness of the companion’s generated responses, as measured by both human survey scores and an “LLM-as-Judge” methodology
- **Video Call (Telecommunication) Performance:** The real-world quality and stability of the WebRTC video communication link.
- **Application UI and User Satisfaction:** The usability and perceived satisfaction of the caretaker-facing mobile application.

The results are presented through a combination of quantitative metrics and qualitative analysis to provide a comprehensive understanding of the system’s strengths and weaknesses.

4.1 Speech-to-Text (STT) Performance

The STT subsystem, utilizing the faster-whisper model, serves as the primary input for the entire interaction pipeline. Its accuracy is therefore critical to the system's overall function, especially given the project's focus on elderly Thai speakers, who may have unique speech patterns, accents, and vocabulary. Performance was evaluated using a

publicly available test dataset of elderly Thai speech assessing both Word Error Rate (WER) and Character Error Rate (CER) as defined in Section 3.6.3.

4.2 Emotion Recognition Performance

Following the transcription, GEMMA 3:4b model was tasked with performing text based emotion classification. The model was evaluated against a manually labeled test set using the metrics outlined in Section 3.6.1. The performance is summarized by the confusion matrix (Figure 4.1) and the per-class metrics (Table 4.2).

4.3 Dialogue Generation Quality

The quality of the HRI (Human Robot Interaction) was evaluated by assessing the dialog generated by the GEMMA 3:4b model, using a human-based survey score and an “LLM-as-Judge” evaluation.

4.3.1 Human Evaluation (Survey Score)

A user study involving 15 participants from the target demographic was conducted. Participants interacted with the “Buddy” system and subsequently rated the quality of its response using a 5-point Mean Opinion Score (MOS) from scale (1=Bad, 5=Excellent). The relevance metrics are Coherence, Fluency, Empathy & Appropriateness, Overall Quality.

4.3.2 LLM-as-Judge Evaluation

To complement the human survey and provide a scalable evaluation, a superior large language model (e.g., GPT-oss or GEMMA 3:27b-it-qat) was employed as a “judge.” The judge was provided with the user’s transcribed input, the detected emotion, and the “Buddy” (GEMMA 3:4b) response. It was then prompted to score the response on a scale of 1-5 across several key criteria.

4.4 Telecommunication system

In the real world situation, the monitoring type app needs to have as much consistency as it could have, especially one that involves the elderly. This section will demonstrate the evaluation on the performance of the telecommunication system. There are 3 factors that need to be considered about

- **Connection success rate:** The percentage of the success attempt over the total number of attempts that occur between two devices.
- **Latency:** The amount of time that two devices need to use for communication.
- **Quality of the call:** The satisfaction that the telecom system can provide to the user.. MOS score, User experience rating

4.5 Satisfaction

The effectiveness of the system also depends on the usability of the caretaker mobile application. A user satisfaction survey was conducted with [N] test users (acting as caretakers) after they performed a set of standardized tasks (e.g., initiating a call, checking an emotion report, viewing the live feed).

A standard System Usability Scale (SUS) survey was used, which results in a score from 0 to 100. Additionally, users were asked to rate specific features.

CHAPTER 5

CONCLUSION

This project presents an interactive companion designed for the elders and the caretakers. Combining an LLMs model, sentiment analysis model, caretaker's application, and elder's application. This system aims to make communication between caretakers and elders easier, especially for elders who struggle with digital devices. By integrating voice activated functions and LLMs on the elder's application, the system reduces the technological barriers and allows a natural interaction between "Buddy" and the elders. On the caretakers side, an elder's emotional state insight is provided, so that caretakers can monitor their beloved emotion, and provide help when needed. Combined with the video call feature, which allows a remote connection.

Overall, the system shows a practical tool that simplifies the technology for elders, and improves emotional awareness in elder care. The combination of this system, brings out the potential of assistive technology to support the aging population, and enhance the quality of remote caregiving.

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